S O P

Standard Operating Procedure

Quick Reference Guide

Airbus A320 Family

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Items marked with a star (*) are to complete also in Turn-Around. A full SOP-Guide with clear and deep explanation in German as well as distinct lessions are availbale via the virtual Aviation Academy (vAA): Dicord: A320_kiddie.

Safety Exterior Inspection

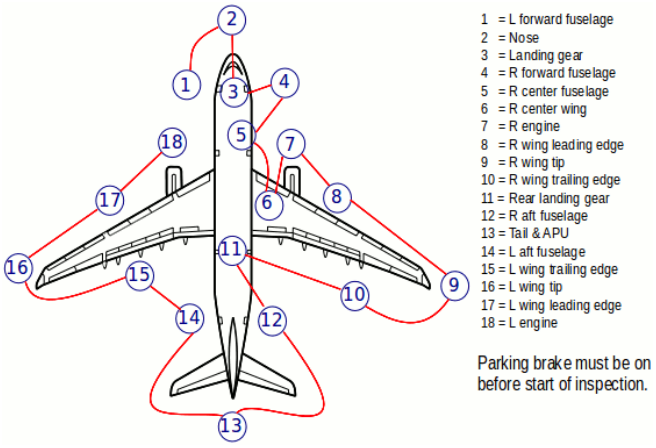
CM1	CM2	
	*Wheel Chocks	IN PLACE
	*Landing Gear Doors	CLOSED
	*APU Area (Inlet Valve and Outlet)	CLEAR

Preliminary Cockpit Preparation

CM1		CM2	
Aircraft Safety Check Flow (Secure for Operation Check)			
		Engine Master Switches 1 & 2	OFF
		Engine Mode Selector	NORMAL
		*Weather Radar System Switch	OFF
		*Weather Radar PWS Switch	OFF
		Thrust Levers	IDLE
		Landing Gear Lever	DOWN
		Windshield Wiper Switches Left & Right	OFF
DC Powerup			
		Battery Voltage BAT 1 & 2 > 25,5V	CHECKED
		Battery 1 & 2 Switch	AUTO
		*External Power	ON
Cockpit Lights			
Overhead Integrated Light	BRIGHT		
Ice Indication & Standby Compass Light	AS RQRD		
Annunciator Lights: Test, set BRT (day) or DIM (night)	CHECKED, SET		
PFD & ND Lights	AS RQRD	PFD & ND Lights	AS RQRD
AFCU and Glareshield Panel Integrated Light	AS RQRD		
Main Panel Flood Light, Pedestal Flood Light, Main Panel Integrated Light	AS RQRD		
ECAM Display Lights	AS RQRD		
CPDLC Brightness	AS RQRD	CPDLC Brightness	AS RQRD
MCDU Brightness	AS5 RQRD	MCDU Brightness	AS RQRD
Dome Light	AS RQRD		
EFB Initialization			
		*ACARS and EFB	INITIALIZED
		*FMGS PRE-INITIALIZATION (AOC-Menu)	ENTERED
Aircraft Acceptance			
*ECAM Status	CHECKED, NORMAL		
*ECAM RCL Pushbutton	PRESS 3sec.		
Technical State of Aircraft (Log)	CHECKED		
*MEL / CDL Items	CHECKED		
Aircraft Configuration	CHECKED		
*OEB Check (Operations Engineering Bulletin)	CHECKED		
Before APU Start			
		Radio Management Panel (RMP) - NAV Light OFF - SELECTED Light OFF - Ground Frequency SET (RMP 1) - Guard Frequency SET (RMP 2)	Checked, ON
		APU Fire Pushbutton Check	IN and GUARDED
		APU Fire Test	PRESS and HOLD

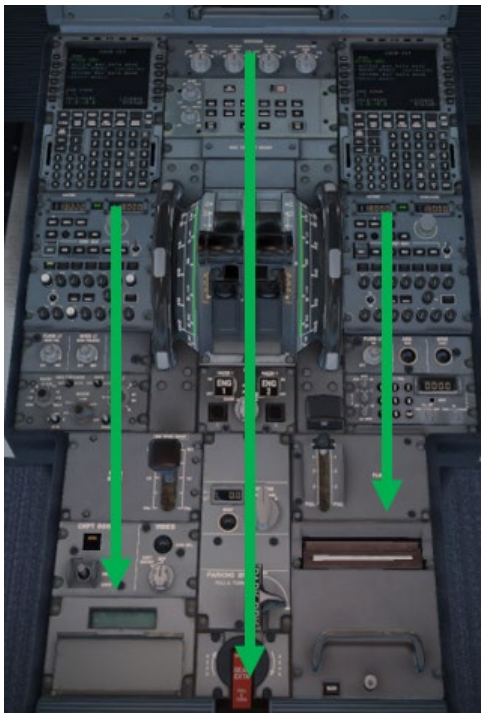
CM1		CM2	
APU Start			
		APU Master Switch	ON
		after waiting for ≥ 3 Seconds or when „FLAP OPEN" Massage on lower ECAM appears:	
		APU Start Selector	START
		ECAM APU Page and EWD: „AVAIL"	CHECKED
		APU Bleed	ON
		Packs 1 & 2 ON	CHECKED
Preliminary Performance Determination			
*Airfield Information (ATIS) incl. METAR	CHECKED	Airfield Information (ATIS) incl. METAR	CHECKED
Takeoff Performance	CALCULATED and CROSSCHECKED	Takeoff Performance	CALCULATED and CROSSCHECKED
Before Walkaround			
		*Oxygen Crew Supply	ON
		*ECAM Door Page: CKPT OXY	verify GREEN, > 1.000psi
		*ECAM HYD Page: Hydraulic Quantity	WITHIN LIMITS
		*ECAM ENG Page: Oil Quantity	SUFFICIENT
		Lower ECAM: All System Data Pages	CHECKED
		*Ground Spoilers (Speedbrakes Lever)	RETRACTED, DISARMED
		Flaps Lever and Flaps Position on EWD	RETRACTED, ZERO
		*Parking Brake	SET ON
		*Brake Pressure Indicators	> 1.000psi
		*Parking Barke Accu Pressure	CHECKED
Emergency Equipment			
		Life Jackets, Smoke-Hoods and Gloves	STOWED
		Protective Breathing Equipment (PBE)	CHECKED
		Portable Fire Extinguisher	STOWED
		Crash Axe	STOWED
		Rear and Overhead Circuit Breaker (CB) Panel (none popped out)	CHECKED
		Maintenance Panel	CHECKED
		Emergency Escape Ropes	CHECKED
		Rain Repellent	CHECKED
		Flash Lights	STOWED
		Oxygen Masks TEST	CHECKED
		*Gear Pins (5pcs) and Covers	ONBOARD and STOWED

Preflight Procedures (Cockpit Preparation)

PF		PM	
Overhead Panel Scan – Flow Pattern from bottom to top and from left to right Principle: „all white lights off“			
*Cockpit Voice Recorder – Ground Control	ON	Aircraft Exterior Inspection	CHECKED
CVR Test (push and hold)	CHECKED		
GPWS Panel	CHECKED		
Emergency Electrial Power Panel	OFF, NO LIGHTS		
Evacuation Panel: Captain	SET		
Flight Control Computers	CHECKED, NORMAL		
*ADIRS Mode Selectors 1, 2 and 3	SET to NAV		
Strobe Lights	AUTO		
NAV & Logo Lights	1 or 2 ON		
*Seat Belt Signs	ON		
No Smoking Signs	AUTO		
Emergency Exit Lights	ARMED		
Wing and Engine 1 & 2 Anti Ice	OFF		
Probe / Window Heat	AUTO / ON		
Cabin Pressurization	AUTO		
Landing Elevation	AUTO		
Cross Bleed Selector	AUTO		
*PACK Flow Selector	AS RQRD[7]		
Temperature Selectors	AS RQRD		
Battery Charger Test	PERFORMED		
Fuel Pumps	ON		
Hydraulics Panel (no Lights)	CHECKED		
Engine 1 & 2 Fire Test	PERFORMED		
Audio Switching Button	NORMAL		
Engine Manual Start Panel	OFF		
Ventilation Panel	AUTO		
Cargo Smoke Panel	AUTO, TESTED		
Cargo Heat and Cargo Vent Panel	SET AS RQRD, NORMAL		
Flight Control Computers	CHECKED, NORMAL		
Radio Management Panel (RMP) 3	VHF 3, DATALINK		
Audio Control Panel (ACP) 3	PA TUNED		

Cockpit Preparation during Walkaround

PF		PM
Main Panel Scan – Flow Pattern in „U-shape“ starting top left		
*Standby ADI (ISIS)	ON, QNH SET	
Terrain on ND	OFF	
*Standby RMI (Radio Magnetic Indicator)	RED FLAGS	
DCDUs	TURNED ON	
Landing Gear Lever	DOWN	
*Parking Brake ACCU Pressure	CHECK GREEN	
*Brake Pressure Indicators	CHECK > 2.000psi	
*Alternate Braking Check	CHECKED	
Terrain on ND	OFF	
*Clock: UTC & Date	CHECKED	
*Anti Skid & Nose Wheel Steering	ON	
Auto Brake	OFF	
Landing Gear Indicators	3 GREEN	
Brake Fan	AS RQRD	

PF		PM
Center Pedestal Panel Scan (from top to bottom and from left to right)		
Radio Management Panel (RMP)	SET	
Audio Control Panel (ACP) VHF 1 CALL Button ON (green illuminated)	ON, VHF ON, Frequencies SET	
Weather Radar System (System OFF, PWS OFF, Gain +4, Tilt +5°, Mode WX+T, Multiscan AUTO, GCS AUTO)	SET for T/O	
Speedbrakes	RETRACTED, DISARMED	
Cockpit Door	NORM, OPEN	
Cockpit Video	OFF	
Display Switching Buttons	NORM	
ECAM Control Panel	CHECKED, SET	
*Thrust Levers	IDLE	
*Engine Master Switches 1 & 2	OFF	
*Engine Mode Selector	NORMAL	
Rudder Trim	0.0	
Parking Brake Handle	SET ON	
Gravity Gear Extension Handle	DOWN & LOCKED	
*Transponder (TCAS Mode AUTO, System 1, TCAS Radar Tilt ABV, Transpnder Mode STBY, SQUAWK 2000)	STANDBY	
Flaps	RETRACTED, ZERO	

FMGS Setup

PF		PM
FMGS Setup („SIF RIP COPY" or „DIFRIPPS")		
Status Page (**D**ata)	SET	
Init A Page	SET	
Flightplan Page	SET	
RAD NAV Page	SET	
Init B Page	SET	
Progress Page	SET, CHECKED	
Performance Page	SET	
COPY Secondary Flightplan	SET	

Cockpit Preparation – Final Flow

PF		PM	
***Glareshield Panel Scan**			
Flight Director (FD)	ON, CYCLE OFF THAN ON	Flight Director (FD)	ON, CYCLE OFF THAN ON
QNH Mode	verify hPa	QNH Mode	verify hPa
QNH	SET	QNH	SET
Altimeter Readings	CROSSCHECKED	Altimeter Readings	CROSSCHECKED
ND Mode	ARC	ND Mode	ARC
ND Range	10NM, SET	ND Range	40NM, SET
VOR / ADF 1 & 2	AS RQRD	VOR / ADF 1 & 2	AS RQRD
LS Mode	OFF	LS Mode	OFF
Constraints (CSTR)	ON	Constraints (CSTR)	ON
FCU Speed	MANAGED	FCU Speed	CROSSCHECKED
FCU Heading	MANAGED	FCU Heading	CROSSCHECKED
FCU Initial Climb Altitude	SET, MANAGED	FCU Initial Climb Altitude	CROSSCHECKED
FMA Modes	CHECK CLB blue, NAV blue	FMA Modes	CHECK CLB blue, NAV blue
ECAM Press Page: LDG ELEV AUTO	CHECKED		
ECAM Status: check for INOP SYS	CHECKED		
ECAM: Fuel on Board (FOB), Balance crosschecked with OFP	CHECKED		

Departure Briefing

PF	PM
Departure Briefing	
	T/O Runway • Runway Identifier • full length or applicable Intersection • Runway Condition (dry / wet)
	SID • Name (Designator) • RNAV Departure or not
	First cleared Altitude (Initial Climb) • xxxxx feet
	MSA (MORA) for the Climb Trajectory • xxxx feet (within xx NM) • (thereafter xxxxx feet)
	Extra Fuel and Time • xx tons / xx minutes
Taxi Briefing • expecting pushback facing xxxxx • thereafter taxi out via xxxxx • to holding point Runway xx • Hotspots in the planned Taxi Route	• For any failure before V1: CM1 will call „STOP" • In case of engine failure after V1: - continue takeoff, no actions before 400ft AGL except gear up - reaching 400ft AGL, ECAM actions - reaching EO ACC altitude, stop ECAM, push for ALT, acceleration and clean up - at green dot: OP CLB, MCT, continue ECAM, after TO C/L, status - ENG OUT routing: EOSID, SID, radar vector, immediate return
Stop Margin for the Rejected Take-Off • xxx meters	
EOSID	
Return / Diversions • In case of any immediate return we are (not) above max landing weight (→ overweight landing checklist QRH)	
Special Operations	
Non-Standard Operations (Non-Normals)	
	Identified Threats
Identified Threats	
Mitigation	
Miscellaneous	

Threats:

AIRPORT	ATC	AIRCRAFT	ENVIRONMENT
• Overload • Construction work • Hotspots • Infrastructure • Runway Condition	• Challenging Restrictions • Language barriers • Phraseology • Short-term changes of Clearances • Similar Callsigns	• MEL OPS • Aircraft Defects • Supplementary Procedures that are not Routine	• Low Visibility • Approach / Runway Lighting • Runway Contamination
WEATHER	**OPERATIONS**	**CREW**	**TERRAIN**
• Windshear • Convective Weather • Cold Weather • Precipitation • Weather Reports, that are not reliable	• Pressure from flight plan or slots (Schedule) • Delays • Crew-Delays • Factors concerning the Load and Balance	• Exhaustion • Low Experience • Negligence • Ablenkung • Distraction • non-standard crew composition	• High Terrain • unknown area • complex Visual Approach

Cockpit Preparation Checklist

♦ COCKPIT PREPARATION CHECKLIST	
GEAR PINS and COVERS	REMOVED
FUEL QUANTITY	_____ KGs, BALANCED
SIGNS	ON and AUTO
ADIRS	NAV
BARO REF	_____ SET [BOTH]

Before Start Procedures

PF		PM
APU Start		
APU Master Switch	ON	
after waiting for ≥ 3 Seconds or when „FLAP OPEN" Message on Lower ECAM appears:		
APU Start Selector	ON	
ECAM APU Page and EWD: „AVAIL"	CHECKED	
APU Bleed	ON	
Packs 1 & 2 ON	CHECKED	
External Power	OFF, DISCONNECTED	

PF		PM	
Before Start Flow Pattern			
Final Loadsheet (AOC Menu)	CHECKED		
Takeoff Performance Data	RECOMPUTE	Takeoff Performance Data	RECOMPUTE
FMGS Takeoff Data	REVISE	FMGS Takeoff Data	CROSSCHECKED
Fuel on Board (FOB)	CROSSCHECKED	Fuel on Board (FOB)	CROSSCHECKED
MCDU: PERF TAKEOFF Page	SELECTED	MCDU: F-PLN Page	SELECTED
ASU and EXTERNAL POWER	DISCONNECTED	Pushback and Startup Clearance (ATC)	OBTAINED
		TCAS Mode	AUTO
		PA: „Cabin Crew, all doors in Flight."	PA ANNOUNCED
Beacon Light	ON	Pushback Truck set for Operations	DONE
Windows, Doors and Slides (check on ECAM Door Page)	CLOSED, ARMED	Windows, Doors and Slides (check on ECAM Door Page)	CLOSED, ARMED
Thrust Levers	IDLE		
Parking Brake ACCU PRESS (green)	CHECKED		
Parking Brake Handle	SET ON		

Before Start Checklist

♦ BEFORE START CHECKLIST	
PARKING BRAKE	_____
TAKEOFF SPEEDS & THRUST	_____ [BOTH]
WINDOWS	CLOSED [BOTH]
BEACON	ON

Pushback Procedure

PF		PM	
Chocks and Ground Equipment	REMOVED	Main Panel Clock	RUN
ACARS	START	Note Off Blocks Time	PERFORMED
NWS Disconnect Memo	CHECKED		
Parking Brake (when asked by ground crew)	OFF		
Pushback	SUPERVISE		
Parking Brake (when pushback completed)	SET ON		

Engine Startup Procedure

PF		PM	
Engine Start Sequence			
Thrust Levers	IDLE		
Dual Cooling	ON		
Engine Mode Selector	IGN/START		
ECAM Engine Page: Ignition Indication (IGN) and Bleed Pressure > 30psi 	CHECKED	Right Engine Area clear	CHECKED
ECAM: no amber crosses & messages	CHECKED		
Announce „Engine Number 2 – Start"	PERFORMED	Engine 2 Master Switch	ON
ECAM: Engine Idle Parameters - N1: 19.5% - N2: 58.5% - EGT: 390°C - FF: 275kg/h - N2 Indication: grey background disappeared - N1 Indication: „AVAIL" ((when Idle-Values reached and stable)	CHECKED	ECAM: Engine Start Sequence - N2 increases: Start Valve inline (open), Bleed Pressure green, Oil Pressure rises - N2 16%: indication of active Ignitor A or B - N2 22%: Fuel Flow (FF) increases (may cross approx. 200kg/h) - EGT & N1: increases within max. 15sec. after FF started (shows that ignition has occurred) - N2 50%: Start Valve closure, Igniter indication off	MONITOR
		Check left Engine Area clear	CHECKED
Announce „Engine Number 1 – Start"	PERFORMED	Engine 1 Master Switch	ON
ECAM: Engine Idle Parameters - N1: 19.5% - N2: 58.5% - EGT: 390°C - FF: 275kg/h - N2 Indication: grey background disappeared - N1 Indication: „AVAIL" (when Idle-Values reached and stable)	CHECKED	ECAM: Engine Start Sequence - N2 increases: Start Valve inline (open), Bleed Pressure green, Oil Pressure rises - N2 16%: indication of active Ignitor A or B - N2 22%: Fuel Flow (FF) increases (may cross approx. 200kg/h) - EGT & N1: increases within max. 15sec. after FF started (shows that ignition has occurred) - N2 50%: Start Valve closure, Igniter indication off	MONITOR
ND Clock Timer	START		
After Start Items (Flow Pattern)			
Dual Cooling	OFF		
Engine Mode Selector	NORMAL	Ground Spoilers	ARMED
APU Bleed	OFF	Rudder Trim	RESET, ZERO
Wing / Engine Anti Ice	AS RQRD	Flaps	SET for T/O
Probe/Window Heat Switch	AUTO	Stabilizer Pitch Trim	SET
APU Master Switch	OFF		
ECAM: STS NORMAL	CHECKED	ECAM: STS NORMAL	CROSSCHECKED
ECAM Memo: **no** „NW STRG DISC"	CHECKED		

Engine Stable Values:

CFM-Engins

- **N1** ~ 20,0% („2")
- **EGT** ~ 400°C („4")
- **N2** ~ 60,0% („6")
- **FF** ~ 300kg/h („3")

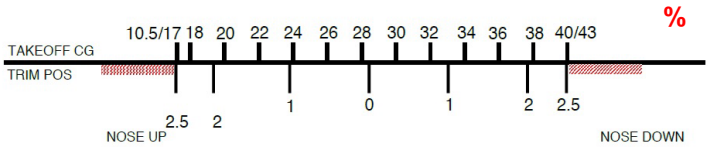
IAE-Engines

N1 ~ 20,0% („2")
EGT ~ 400°C („4")
N2 ~ 60,0% („6")
FF ~ 400kg/h („4")

After Start Checklist

♦ AFTER START CHECKLIST	
ANTI-ICE	_____
ECAM STATUS	CHECKED
PITCH TRIM	_____ % SET
RUDDER TRIM	NEUTRAL

TAKEOFF CG / TRIM POSITION



Manual Engine Start

PF		PM	
Engine Manual Start			
Thrust Levers	IDLE		
Engine Mode Selector	NORMAL		
then: Engine Mode Selector	IGN/START		
No amber crosses & messages	CHECKED		
Engine Manual Start Pushbutton 2	**ON**	Start Valve inline (open)	CHECKED
		N2 increases	CHECKED
		Oil Pressure increases	CHECKED
		N2 reaches 22%[1]	CHECKED
„Engine Number 2 – Start"	ANNOUNCE	**Engine 2 Master Switch**	**ON**
		Chrono (Elapsed Timer)	START
		Indication of active Ignitor A or B	CHECKED
		Fuel Flow (FF) increase	CHECKED
		max. 15sec. after Fuel Flow starts:	
		EGT increase	CHECKED
		N1 increase	CHECKED
		N2 reaches 50%	CHECKED
		Start Valve crossline (closed)	CHECKED
		Ignitor Indication OFF	CHECKED
		Engine Idle Parameters	NORMAL
		Chrono (Elapsed Timer)	STOP
Engine Manual Start Pushbutton 2	**OFF**		
Engine Mode Selector	NORMAL		
then: Engine Mode Selector	IGN/START		
No amber crosses & messages	CHECKED		
Engine Manual Start Pushbutton 1	**ON**	Start Valve inline (open)	CHECKED
		N2 increases	CHECKED
		Oil Pressure increases	CHECKED
		N2 reaches 22%[1]	CHECKED
„Engine Number 1 – Start"	ANNOUNCE	**Engine 1 Master Switch**	**ON**
		Chrono (Elapsed Timer)	START
		Indication of active Ignitor A or B	CHECKED
		Fuel Flow (FF) increase	CHECKED
		max. 15sec. after Fuel Flow starts:	
		EGT increase	CHECKED
		N1 increase	CHECKED
		N2 reaches 50%	CHECKED
		Start Valve crossline (closed)	CHECKED
		Ignitor Indication OFF	CHECKED
		Engine Idle Parameters	NORMAL
		Chrono (Elapsed Timer)	STOP
Engine Manual Start Pushbutton 1	**OFF**		
Engine Mode Selector	NORMAL		

Taxi Flow Pattern

PF		PM	
Taxi Items (Flow)			
Flight Controls	CHECKED	Flight Controls	CHECKED
LS Pushbutton	ON	LS Pushbutton	ON
		Taxi Clearance	OBTAINED
Area Clearance: „Left Side clear"	ANNOUNCED	Area Clearance: „Right Side clear"	ANNOUNCED
Nose Gear Lights	TAXI		
Runway Turnoff Lights	ON		
Parking Brake	OFF		
Thrust Levers	AS RQRD		
Brake Pedals (decelerate)	PRESSED	Check Brake Pressure	ZERO
Call „Brake Check"	ANNOUNCED	Call „Pressure Zero"	ANNOUNCED
		ATC Clearance Confirmation (silent)	
		F-PLN Page: Departure Runway	CONFIRM
		F-PLN Page: SID	CONFIRM
		FCU: Initial Climb Altitude	CONFIRM
		EFIS: Both FDs ON	CONFIRM
Departure Briefing Confirmation			
„No changes" or Update	ANNOUNCED		
		Taxi Flow Pattern	
		Autobrakes	MAX
		Terrain on ND (only on PM side)	ON
		SQUAWK	CONFIRM / SET
		Transponder: ABV selected, TCAS Mode ON	CONFIRM
		Weather Radar: System 1 ON, PWS AUTO	SET
		ECAM: TO CONFIG Pushbutton	NORMAL
		ECAM: Memo Takeoff NO BLUE	CHECKED
„Taxi Checklist"	ANNOUNCED		
Taxi Checklist	READ	Taxi Checklist	READ

Taxi Checklist

♦ TAXI CHECKLIST	
FLIGHT CONTROLS	CHECKED [BOTH]
FLAPS SETTING	CONFIG _____ [BOTH]
RADAR & PRED WINDSHEAR	ON and AUTO
ENG MODE SELECTOR	_____
ECAM MEMO AUTO BRK MAX SIGNS ON CABIN READY SPLRS ARM FLAPS TO TO CONFIG NORM	TAKEOFF NO BLUE
CABIN	READY

Before Takeoff Flow Pattern

PF		PM	
Before Takeoff Scan Flow			
ND Clock Timer (verify > 3mins warmup)	RESET	Brakes Temperature < 150°C	CHECKED
		Brake Fans	OFF
		Takeoff / Lineup Clearance	OBTAINED
		TO SHIFT	ADJUST
Approach Path clear of Traffic	CHECKED		
PA: „Cabin Crew, prepare for Departure."	PA ANNOUNCED		
Strobe Lights	ON		
		Packs 1 & 2	AS RQRD
		Transponder	TA/RA
Sliding Tables	STOWED	Sliding Tables	STOWED
„Line-Up Checklist"	ANNOUNCED		
Line-Up Checklist	READ	Line-Up Checklist	READ
Nose Gear Lights	TAKEOFF		
Landing Light Left & Right	ON		

♦ LINE-UP CHECKLIST	
TAKEOFF RWY	_____ [BOTH]
TCAS	_____
PACKS 1 & 2	_____

Takeoff and Initial Climb

PF		PM	
Takeoff Procedure			
Call: „Takeoff"	ANNOUNCE	Elapsed Timer and Chrono	START
Thrust Levers	SET 50% N1	Thrust stabilized (N1 at 50% on both engines)	CHECKED, callout: „Stabilized"
Sidestick Forward Input	SET		
Brakes	RELEASED		
Takeoff Thrust	SET FLEX / TOGA	PFD and ECAM Engine Indication	MONITOR
FMA: „MAN Flex XX, SRS, Runway, A-Thrust blue"	ANNOUNCE	Call: „Thrust Set"	ANNOUNCE
at 80kts: slowly reduce Sidestick Pressure	RELEASE		
		at 100kts call „100 knots"	ANNOUNCE
at 100kt (when announced), call „checked"	CHECKED		
		at V1 call „V1"	ANNOUNCE
Hand von den Thrust Lever nehmen	PERFORM		
		at VR call „Rotate"	ANNOUNCE
Nose Pitch Up to 15°	INCREASE		
		call „Positive Climb"	ANNOUNCE
Order: „Gear Up"	ANNOUNCE	Landing Gear Lever	UP
		Call „Gear Up"	ANNOUNCE
FMA: „NAV"	ANNOUNCE		
Autopilot	ON		
at Thrust Reduction Altitude			
Thrust Levers	CLIMB (CL-Raste)	Packs 1 & 2 (mit 10sec. Abstand!)	ON
FMA: „Thrust CLIMB, CLIMB, Autothrust"	ANNOUNCE		
Order: „Flaps 1" (at F-Speed with positive speed trend)	ANNOUNCED	Call: „Speed checked"	ANNOUNCED
		Flaps 1	SELECT
		confirm: „Flaps 1"	ANNOUNCED
Order: „Flaps 0" (at **S-Speed** with positive speed trend)	ANNOUNCED	Call: „Speed checked"	ANNOUNCED
		Flaps 0	SELECT
		Confirm: „Flaps 0"	ANNOUNCED

PF		PM	
Acceleration Flow Pattern			
at Acceleration Altitude			
Ground Spoilers (Speedbrakes)	DISARMED	Nose Gear Lights	OFF
		Runway Turnoff Lights	OFF
at Transition Altitude			
Order: „Set Standard“	ANNOUNCED		
Baro Ref	PULL STD	Baro Ref	PULL STD
		Confirm: „Standard set, cross-checked, passing FL ___ … now.“	ANNOUNCED
Altitude	CHECKED		
Confirm: „checked“	ANNOUNCED		
		Acceleration Flow Checks	
		Flaps	RETRACTED, ZERO
		Spoilers	RETRACTED, DISARMED
		Engine Mode Selector	AS RQRD
		Landing Gear	UP, NO LIGHTS
		Packs 1 & 2	CHECKED ON

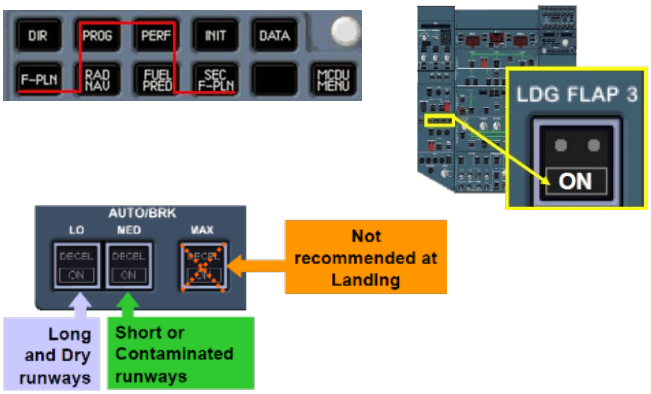
Climb

PF		PM	
Climb Procedure			
passing 10.000ft („10-Checks“)			
Landing Lights	OFF, RETRACTED		
Seat Belt Signs	AS RQRD		
„Cabin Crew released“	ANNOUNCED		
EFIS Options			
LS Pushbutton	OFF	LS Pushbutton	OFF
ARPT (or CSTR)	ON	ARPT (or CSTR)	ON
TERR on ND	OFF	TERR on ND	OFF
ND Range	SET	ND Range	SET
Upper ECAM: Engine Parameters	CHECK STABLE		
ND Chrono (verify < 10min. TOGA)	STOP and CLEAR		
ECAM Cruise: Pressurization	CHECKED		
ECAM Memo	REVIEWED		
Weather Radar Tilt	–2°		
Anti Ice	AS RQRD		
MCDU: „Clear and Copy“			
		FIX INFO: clear Fix Info	CLEAR
		RAD NAV: clear tuned Navaids	CLEAR
		PROG Page	CHECKED, SET
		SEC F-PLN	COPY ACTIVE
passing FL 200			
Weather Radar Gain	CAL		
ECAM Cruise: Pressurization	CHECKED		
Passenger Announcement	ANNOUNCED		
passing FL 300			
ECAM Cruise: Pressurization	CHECKED		

Cruise

PF		PM	
Cruise Items			
FMA: „ALT CRZ“ on Top of Climb	CHECKED	**Periodic ECAM Checks** (about every 30 minutes)	
FCU Altitude	SET & MANAGE	ECAM Memo	REVIEWED
TCAS Radar Tilt	BELOW	ECAM System Data Pages	REVIEWED
Weather Radar Tilt	–1°	ACARS Messages	REVIEWED
Adjust ND Range as appropriate	SET		
		Flight Progress Monitoring (repeatedly)	
		Flight Progress (Track & Distance)	CHECKED
		MCDU PROG Page: Nav Accuracy	CHECKED
		Fuel Checks (at least every 30 minutes)	
		Fuel Consumption	CHECKED
		Fuel Leakage	CHECKED

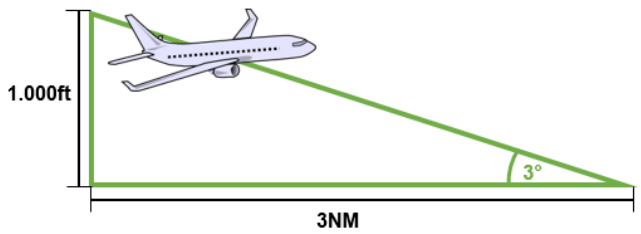
Descent Preparation

PF		PM	
Descent Preparation (latest about 80NM prior to Top of Descent ~ 10 Minutes)			
ECAM Status Page: any abnormality?	CHECKED		
ECAM Cruise Page: LDG ELEV AUTO	CHECKED		
Weather & Landing Information (ATIS)	OBTAINED		
Landing Performance Calculation - Landing Distance required / available	CALCULATED		
FMGS Preparation: Ω - F-PLAN Page - RAD NAV Page - PROG Page - PERF Page - FUEL PRED Page - SEC F-LPN Page	PERFORMED		
GPWS LDG FLAP 3	AS RQRD		
Auto Brake	AS RQRD		
Cleared Descent Altitude on FCU	SET	Descent Clearance	OBTAINED

Approach Briefing

Merke:

1. **Altitude to be reduced x 3 = Distance needed for the Descent** (PROG Page!)
2. **Altitude at which you should currently be = Milage to Destination x 3** (PROG Page!)
3. **Sinkrate = GS : 2 x 10**

PF	PM
Arrival Briefing	
	MORA/MOCA/MSA • We are approaching _____ from the [Direction] out of [CRZ FL] • The MSA of our Final for Runway [RWY] is xxxxx feet and xxxx feet within xx NM • and in case of a missed approach it's xxxx feet.
	STAR • We are expecting the [STAR Name] arrival for Runway [RWY] • via Transition [Name]. Our Clearance-Limit is [WAYPOINT].
	Type of Approach • We are expecting an [ILS] Approach for Runway [RWY]
	Minimums • Minimum is xxx feet, • Baro xxx set
	Go Around Trajectory • In case of Go Around we climb [Routing according to ILS Chart]
	Extra Fuel and Time • We have xx tons and xx minutes of hold.
Guidance for the Approach • Our Guidance for the Approach is gonna be LOC (Localizer) and G/S (Glideslope) | NAV/FPA | Final APP, etc.	
Landing Flap Setting • We are using Config Flaps [FULL].	
Stop Margin • is xxx feet (according to Landing Performance Calculation)	
Use of Reverse • We're gonna use Max [IDLE] Reverse Thrust	
Use of Autobrake • Autobrake is LOW [MEDIUM]	
Planned Runway Exit • We plan to vacate to the right [left] via [EXIT], • then right [left] on [TWY] … • and follow ATC instructions.	
Hotspots for Taxi In • We have [Number] hotspots in the way, one is [Identifier] and [Identifier] junction.	
Special Operations • Long Taxiway • Single Engine Taxi In (SETI)	
Non-Standard Operations • No Non-Standard Operations. [INOP Systems in der Status Page]	
	Identified Threats
Identified Threats	
Mitigation	
Miscellaneous	

Descent

PF		PM	
Descent Items			
MCDU: PERF Page	SET	MCDU: F-PLN Page	SET
		Cabin Crew	ADVISED
FCU Altitude	(OPEN) DESCENT		
FMA: „THR IDLE, OPEN DES, NAV“	CALLOUT	Descent Progress	MONITOR
Speedbrakes	AS RQRD		
Engine Anti Ice	AS RQRD		
Wing Anti Ice	AS RQRD		
Weather Radar Tilt	ADJUSTED		
Passing FL 300			
ECAM Cruise Page: Pressurization	CHECKED		
Weather Radar Tilt	ADJUSTED		
Passing FL 200			
ECAM Cruise Page: Pressurization	CHECKED		
Weather Radar Tilt	ADJUSTED		
Passing FL 150			
Seatbelt Signs	ON		
Passing FL 100 Procedures („10.000ft Gate”)			
Landing Lights	ON		
Seatbelt Signs	CHECK ON		
PFD Speed: IAS ≤ 250kt	CHECKED		
PA: „Cabin Crew, prepare for Landing”	PA ANNOUNCED		
EFIS: CONSTRAINTS	SELECTED	EFIS: CONSTRAINTS	SELECTED
EFIS: ND Range	SELECTED	EFIS: ND Range	SELECTED
LS Pushbutton	AS RQRD	LS Pushbutton	AS RQRD
		Terrain on ND	ON
RAD NAV Page: VOR & Course selected	SET		
PROG Page: GPS PRIMARY	CHECKED		
PROG PAGE: Navigation Accuracy HIGH	CHECKED		
SEC F-PLN	COPY ACTIVE		
ECAM Cruise Page: Pressurization	CHECKED		
ENGINE MODE Selector	NORM/IGN		
at Transition Level			
Baro Ref	Local QNH SET, CROSSCHECKED	Baro Ref	Local QNH SET, CROSSCHECKED
„Approach Checklist“	ANNOUNCED		
Approach Checklist	READ	Taxi Checklist	READ

♦ APPROACH CHECKLIST	
BARO REF	_____ SET [BOTH]
SEATBELT SIGNS	ON
MINIMUM	_____
AUTO BRAKE	_____
ENG MODE SELECTOR	_____

If you enter the approach “hot and high”, i.e. too high and too fast, and the distance to the runway treshold (route) is no longer sufficient to reduce the required altitude by conventional means, we recommend the following procedure according to “Descent per Mile”
:

- **Extend speedbrake** (possibly also briefly autopilot OFF in order to be able to extend the speedbrakes FULL)
- **Level off** (press the vertical speed selector on the FCU: Push to Level off)
- **reduce Speed** (**Selected Speed** down to **Clean Speed** or, if you are flying below 220kt, to 180kt). To fly at minimum clean speed (O-speed; “green dot” speed), you can activate the approach phase in the MCDU, pull the speed selector on the MCP and turn the selected airspeed down to the green dot speed displayed on the speed tape of the PFD..
- when the desired altitude is reached: **OPEN Descent**
- <u>if you are already on the **final approach**, additionally::</u>
- **Flaps 2**
- **Gear DOWN**
- **OPEN Descent**

This “Descent per Mile” procedure means you fly slower, but achieve a higher descent rate over the distance.

Initial Approach Data (summary and overview)

Recommended Speed Gates:

- ≤ 10.000ft … IAS < 250kts
- from End of STAR … IAS max. 210kts
- till 10nm to Touchdown … IAS max. 180kts
- till 4nm to Touchdown … IAS max. 160kts
- on Glideslope … IAS max. 180kts
- V_{APP} … ca. 140kts IAS

DME	IAS
12NM	220kts
11NM	210kts
10NM	200kts
9NM	190kts
8NM	180kts
7NM	170kts
6NM	160kts
5NM	150kts
4NM	140kts

Altitude:

- Distance to RWY-Treshold [NM] x 3 = Flightlevel
  The vertical deviation (VDEV) can be checked on the Progress Page (PROG) of the MCDU. The LSK2R line shows the current vertical deviation to the planned descent profile in ft, whereby a plus symbol (+) means that you are flying too high and a minus symbol (-) means that you are already too low.

Sinkrate:

- Sinkrate [ft/min.] for –3° = GS x 5 (= GS x 10 : 2)
  The required descent rate, which must be flown from the final approach fix (FAF) in feet per minute for a descent at 3°, as is the case in most approaches, can be estimated very simply by multiplying the groundspeed (GS) by 5 (GS x 5 or GS x 10 : 2). As a rule, this is about 700ft/min. With this descent rate you fly down to an altitude of about 30 feet above the touchdown zone, then from 30ft radar altitude you slowly raise the nose and take the thrust out, i.e. to idle thrust.
  In principle, a descent rate of up to 7,500ft/min. is possible without danger; the standard descent rate is 2,000 - 5,000ft/min. Again, it is helpful to calculate the sink rate (vertical speed) using the rule of thumb GS x 5. In descent, **either 10kts speed or 300ft altitude can be reduced within 1NM**.

Flap Settings:

Position	Max. Flaps/Slats Speeds (V_{FE})	Slats	Flaps	Flight Phases	
0		0°	0°		Cruise / Hold
1	**238kt**	18°	0°		Hold / Approach
	285kt		10°	Take off	Ground (< 100kt)
2	**215kt**	22°	15°		Approach
3	**195kt**	22°	20°		Approach / Landing
Full	**186kt**	27°	35°		Landing

Final Approach and Landing – Instrument Landing System (ILS)

PF		PM	
ILS Approach			
Descent to Final Approach Altitude, prior to Intercepting the Localizer (LOC)			
Selected Speed in Clean Configuration	220kt		
LS Pushbuttons	ON		
APPR Mode (FCU)	ARMED		
Second AP	ENGAGED		
Approach Phase (MCDU)	ACTIVE		
ILS-Approach Indication (ND)	CHECKED		
ILS Frequency and Distance (PFD)	CHECKED		
LOC und G/S blue (FMA)	CHECKED	Weather Radar Tilt	+4°
Altitude	CHECKED	PFD: Vertical Profile	MONITOR
when Localizer Intercept			
FMA: LOC* → LOC green	MONITOR		
Selected Speed: decelerate	180kt		
Callout: „Flaps 1“	ANNOUNCED	Callout: „Speed checked“	ANNOUNCED
		Flaps 1	SET
Deceleration towards S-Speed	CHECKED	Callout: „Flaps 1“	ANNOUNCED
prior to Glideslope Capture			
Callout: „Flaps 2“	ANNOUNCED	Callout: „Speed checked“	ANNOUNCED
		Flaps 2	SET
Deceleration towards F-Speed	CHECKED	Callout: „Flaps 2“	ANNOUNCED
FMA: G/S* → G/S green	MONITOR		
FCU A/THR Speed Mode	MANAGED		
Order: „Set Go Around Altitude ____ft“	ANNOUNCED	Go Around Altitude	SET
		Confirm: „Go Around Altitude ____ft set“	ANNOUNCED
at about 2.000ft RA or 6nm inbound ILS – configure the aircraft:			
Order: „Gear down“	ANNOUNCED	Landing Gear Lever	DOWN
		Confirm: „Gear down“	ANNOUNCED
Callout: „Flaps 3“	ANNOUNCED	Callout: „Speed checked“	ANNOUNCED
		Flaps 3	SET
Deceleration towards V_{APP}	CHECKED	Callout: „Flaps 3“	ANNOUNCED
Callout: „Flaps Full“	ANNOUNCED	Callout: „Speed checked“	ANNOUNCED
		Flaps Full	SET
		Callout: „Flaps Full“	ANNOUNCED
		AutoBrakes	CONFIRM
		Ground Spoilers	ARMED
		Runway Turnoff Lights	ON
		Nose Gear Lights	TAKEOFF
		Wing Anti Ice	OFF
		Cabin Crew (Purser Call Button)	ADVISED
so we are „stable“ in Final Approach („fully configured”)			
Landing Checklist	READ		
1.000ft Gate (RA)			
Announce: „1.000, STABILIZED“ or: „Go Around, Flaps“	ANNOUNCED		
Runway in Sight			
Autopilot	DISENGAGE		
Flight-Directors	AS RQRD		
A/THR	AS RQRD		
Thrust Levers	RETARD to N1		
Callout: „Autopilot OFF, manual Flight, I have control”[17]	ANNOUNCED		

PF		PM	
ILS Approach			
at MDA + 100ft			
		Announce: „Hundred Above“	ANNOUNCED
Replay: „Checked“	ANNOUNCED		
at Decision Altitude (MDA; „Minimum“)			
		Announce: „Minimum“	ANNOUNCED
Announce: • „Continue“ (Runway in sight and stable Approach) or • „Go Around, Flaps“	ANNOUNCED		
from about 200ft AGL (RA)			
only look outside	AIMING POINT		
bei etwa 100ft AGL (RA)			
Gaze fixed only on the aiming point	AIMING POINT		
50ft AGL (RA)			
Gaze wanders up	END OF RUNWAY		
Across Runway Treshold	TRACK		
Ready for Flare	PREPARED		
shortly before 30ft AGL			
slight pull on the sidestick	APPLY		
Flare and **Thrust Idle** *if the approach is too high*	APPLY		
30ft AGL			
View to the end of the runway	END OF RUNWAY	Engine Parameters	MONITOR
Flare: 2 – 3° to max. 7.5°	NOSE PITCH UP		
Thrust Levers	RETARD TO IDLE		
Crosswind-Compensation	APPLY RUDDER		
Touch-Down			
Sidestick-Input	RELEASE		
Nose Pitch Derotation	MONITOR		
Reverser Thrust	IDLE / FULL		
		Announce: „Ground Spoilers“	ANNOUNCED
		Announce: „Reverse Green“	ANNOUNCED
		Announce: „Decel“[25]	ANNOUNCED
Rollout	APPLY RUDDER		
		Announce: „70 knots“	ANNOUNCED
„No Actions on the Runway!“[27]			
at 70kts GS			
Reverser Thrust	IDLE		
Toe Brakes	ACTION		
Announce: „manual braking“	ANNOUNCED		
at 30kts GS (Taxi-Speed)			
Thrust Levers	FORWARD IDLE		
Auto-Brakes	OFF		
Vacate Runway on next Exit-Point	EXIT RWY		

♦ LANDING CHECKLIST	
MISSED APPROACH ALTITUDE	_____ft SET
ECAM MEMO LDG GEAR DN SIGNS ON CABIN READY SPLRS ARM FLAPS SET	LANDING NO BLUE
CABIN	READY

1.000ft Gate Scan:

- **PFD**: - **on Localizer** centered (1)
  - **on Glideslope** centered (2)
  - **on Target Speed** V_{APP} (3)
- **Upper ECAM**: - **N_1 Spoolup** (4)
  - **Flaps FULL** (5)
  - **Landing Checklist completed** (6)

→ if not: **GO AROUND!**

PF		PM	
Discontinued Approach			
1.000ft AGL (RA)			
		Announce: „1.000, UNSTABLE“	ANNOUNCED
Announce: „CANCEL APPROACH“	ANNOUNCED		
FCU Approach (or LOC) Pushbutton	DISARMED		
FCU Lateral Mode: NAV / HDG	SET		
FCU Vertical Mode: V/S or LEVEL OFF	SET		
FCU Speed: select and adjust	SET		

PF		PM	
Go Around (Missed Approach Procedure)			
Decision at MDA			
Order: „Go Around, Flaps“	ANNOUNCED	Confirm: „Speed checked, Flaps 3“ or „Speed checked, Flaps 2“	ANNOUNCED
Thrust Levers	TOGA		
Rotate 15° and follow FD (SRS)	PULL	Flaps Lever rectract one step	SET
FMA: „MAN TOGA, SRS, Go Around Track, A-Thrust blue“	ANNOUNCED		
When Positive Climb			
		Confirm: „Positive Climb“ (initially 15° Nose Pitch Up)	ANNOUNCED
Order: „Gear Up“	ANNOUNCED		
		Confirm: „Gear Up“	ANNOUNCED
		Landing Gear	UP
		Ground Spoilers	DISARMED
		Nose Gear Lights	OFF
		Runway Turnoff Lights	OFF
FCU: HDG or NAV Mode Minimum: 100ft AGL	ENGAGE		
Confirm: „Heading“ or „NAV“	ANNOUNCED		
Autopilot	ON		
at Thrust Reduction Altitude			
Thrust Levers when „LVR CLB“ flashing on FMA	CL		
at Acceleration Altitude			
FCU Traget Speed: Green Dot Speed	MANAGED		
at F-Speed with positive speed trend: Order: „Flaps 1“	ANNOUNCED	Callout: „Speed checked“	ANNOUNCED
		Flaps 1	SET
		Callout: „Flaps 1“	ANNOUNCED
at S-Speed with positive speed trend: Order: „Flaps 0“	ANNOUNCED	Callout: „Speed checked“	ANNOUNCED
		Flaps 0	SET
		Callout: „Flaps 0“	ANNOUNCED
Continue on Standard Missed Apporach Procedure and Enter Holding Pattern			

Approach Types

Precision Approaches	Non-Precision Approaches
ILS	LOC
MLS	VOR
PAR	NDB
GLS	RNAV LNAV

APV-Approaches
FLS
RNP LNAV/VNAV
LPV

						Guidance Modes		
Approach	**LS-Pb**	**V/DEV**	**APPR**	**LOC**	**TRK/FPA**	**lateral Guidance (Heading)**	**vertical Guidance (Descent)**	**Autopilot disconnect**
ILS	✓	✗	✓	✗	✗	Managed (LOC)	Managed (G/S)	CAT I: 160ft CAT II: 80ft CAT III: 0ft
FLS	✓	✗	✓	✗	✗	Managed (F-LOC)	Managed (F-G/S)	0ft AGL
RNAV-LNAV	✗	✓	✓	✗	✓	Managed (NAV)	Managed (FINAL APP) Selected (FPA)	250ft AGL 250ft AGL
RNP LNAV/VNAV	✗	✓	✓	✗	✗	Managed (NAV)	Managed (FINAL APP)	250ft AGL
LPV	✗	✓	✓	✗	✗	Not Authorized		
LOC	✓	✗	✗	✓	✓	Managed (LOC) Selected (TRK)	Selected (FPA)	250ft AGL
VOR/NDB	✗	✓	✓	✗	✓	Managed (NAV) Selected (TRK)	Managed (FINAL APP) Selected (FPA)	250ft AGL 250ft AGL

The **LS pushbutton** is always switched on (**ON**) for the following approach types that offer vertical guidance:

- **ILS-Approach**
- **LOC-Approach**
- **FLS-Approach**

An **RNAV approach**, on the other hand, is flown with the vertical deviation function of the FMGS (V/DEV), just like a **VOR approach** or an **NDB approach**, which is displayed as a "**green brick**" on the **vertical deviation scale** of the PFD. The **LS pushbutton** is therefore left **switched off** for these approaches.

The **approach mode** of the FCU can always be switched on, except for a **localizer** (**LOC**) **approach**. Here, however, the **LOC mode** is used.

The **guidance modes** show what can be flown and how, i.e. managed in both areas, managed and selected or selected in both areas. An RNAV approach must never be flown with TRK/FPA all the time, so that you are following a selected track. The lateral guidance must be provided by the FMGS itself, but the altitude can be set to selected and so can the FPA. An RNP LNAV/VNAV approach, on the other hand, may only be flown with Final APP, i.e. laterally and vertically managed.

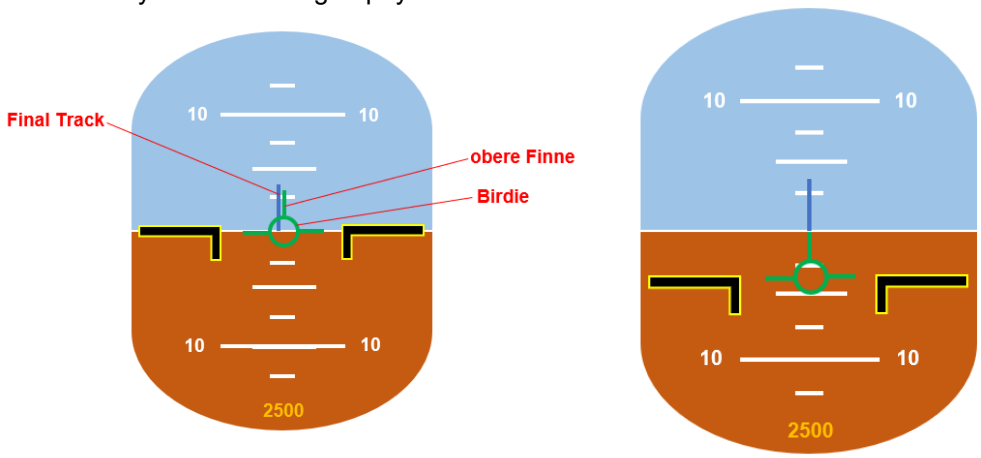
It is generally recommended for every approach to disengage the autopilot (AP) at 200ft AGL (above airfield elevation) at the latest and then fly the last section of the approach visually, unless an ILS autoland is planned. An autoland is only possible with an ILS approach if both autopilots are active in APPR mode and the FMA displays a CAT II or CAT III capability. In all other cases, the autopilot system switches off automatically at a certain altitude, depending on the autoflight mode.

In principle, every approach is flown with the flight director (FD) ON, but in manual flight, i.e. after the autopilot (AP) is disengaged, the flight director (FD) can be switched off. In an RNAV or RNP approach without autopilot, the flight director (FD) must be switched off. When using FPA in particular, it is recommended that the flight director is switched off, as otherwise the birdie of the TRK/FPA mode is not displayed in a meaningful way so that it can be followed:

→ in **Manual Flight** (**optional**), in **TRK-FPA** (**recommended**) and **RNAV-** or **RNP-Approach in Manual Flight** (**mandatory**):

- **Autopliot OFF**
- **Flight Director OFF**
- **Final Track SET**

In TRK/FPA mode, “flying after the bird” means placing the upper fin of the bird exactly on the horizon line in order to descend in a 3.0° glidepath. The heading selector on the FCU is used to turn in the final track, resulting in a small, blue, vertical line on the Attitude Indicator, which then only needs to be kept centered. However, the radial must first be intercepted for a VOR approach, otherwise you will fly parallel to the approach radial on the final track. Then, however, only the green fin of the birdie pointing upwards needs to be held on the blue vertical track line, which means that the final track can be flown exactly without having to pay extra attention to the wind conditions.



In daily operational practice, an ILS is usually flown, in the last section of the final from 200ft AGL with autopilot OFF and flight directors ON.

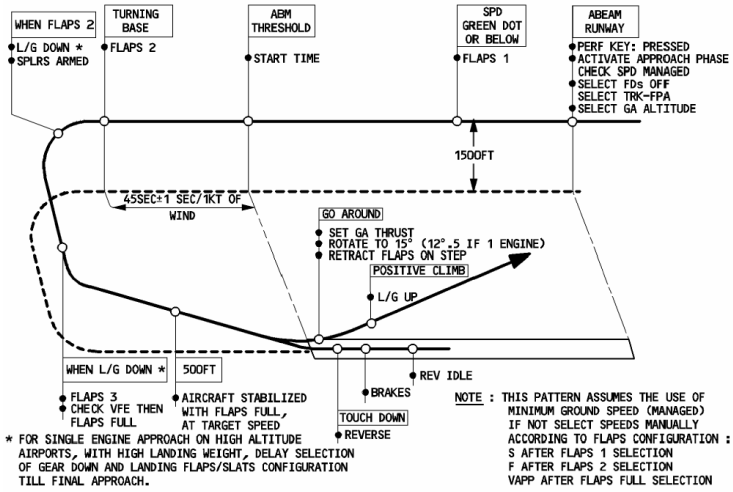
Visual Swingover and Visual Approach in practice

A visual swingover, which is quite common on the approach to EDDF (Frankfurt), is basically flown like any other visual approach. It is particularly important to remember that both flight directors (FD) must be switched OFF, as they are set to the originally planned arrival runway and would therefore provide incorrect displays for the runway on which a visual swing is to be flown. In this context, it is also important to remember that the Go Around is also programmed for the originally planned arrival runway. Depending on which go-around procedure is programmed, it is important to observe the FMA:

- „**GA TRACK**" (**Go Around Track**) means that the go around is simply flown straight ahead. In this case, the go-around programmed for the originally planned runway is irrelevant.
- However, if „**NAV**" is displayed in the FMA, the FMGS flies the missed approach procedure of the originally planned and programmed runway, which would now be incorrect and could also lead to a conflict with an aircraft in the go-around of the other runway. In this case, the heading selector on the FCU must therefore be pulled in order to first maintain the runway heading and then fly the corresponding missed approach procedure on the new runway.

Approach relatively low (1,500ft AGL) and relatively slow (Green Dot Speed or below) with FLAPS Config 1 in the downwind segment:

- **FD OFF** (both)
- **Autopilot OFF**
- **Autothrust** (A/THR) **OFF**
- **G/S Mode OFF** (on GPWS-Panel of the Overhead-Panel)
- **Approach relatively low (1,500ft AGL) and relatively slow (Green Dot Speed or below) with FLAPS Config 1 in the downwind segment**
- When **abeam Runway-Treshold**:
  - Elapsed Timer START: **continue straight ahead for 3 seconds per 100ft of necessary altitude reduction +1 second per knot of headwind or –1 second per knot of tailwind**
- **After the time has elapsed** (usually just **45 Seconds**):
  - **FLAPS 2**
  - **Turn left by 90°**
  - **Landing Gear DOWN**
  - **Spoilers ARMED**
  - Descent with **300ft/min. Descent-Rate**
- When **abeam Runway**:
  - **FLAPS 3**
  - **Turn left by 90°**
  - Speed: **V_{REF}** (MANAGED Speed)
- When **Landing is assured**:
  - **FLAPS FULL**

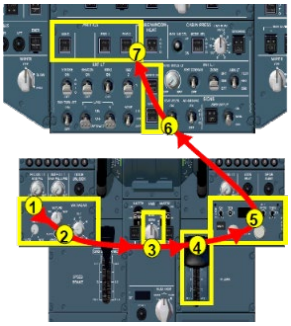


The **actual visual approach** then usually begins **3 or 4 miles final**, i.e. 3 - 4nm inbound to the runway treshold. It is therefore advisable to set a **4nm circle around the runway treshold** via the **FIX INFO page** of the MCDU. To do this, enter the destination airport as a fix, followed by the runway designator of the Arrival_Runway and assign it a radius of 4nm. At this point, the **final turn** should be completed and the aircraft should already be flying **wings level** again, on **final track** with 1,000ft AGL altitude and "**fully configured**".

- **4 Miles Final:**
  - **Wings Level**
  - **on Final Track**
  - Altitude: **1.000ft AGL**
  - „**fully configured**": Landing Gear DOWN, FLAPS Final (Config FULL or Config 3), Spoilers ARMED

The final approach itself up to landing is then carried out according to **visual cues** (runway aiming point) and **PAPI lights**. The procedure described for the visual approach is primarily used for energy management in order to be able to reduce speed quickly.

After Landing Flow Pattern

PF		PM	
After Landing Items			
Flight Directors (FD)	OFF	Flight Directors (FD)	OFF
Landing Scales (LS)	OFF	Landing Scales (LS)	OFF
Ground Spoilers	DISARMED	Clock Timer	STOP
Landing Lights	OFF, RETRACED	Weather Radar	OFF
Strobe Lights	AUTO / OFF	Predictive Windshear (PWS)	OFF
Nose Gear Lights	TAXI	Engine Mode Selector	NORMAL
		Flaps	RETRACTED, 0
		TCAS Mode Selector	AUTO
		Transponder	STANDBY, SQUAWK 2000
		APU Master Switch	ON
		APU Start Selector (3 Sekunden nach dem APU Master Switch)	ON
		APU Bleed	ON
		Wing and Engine Anti Ice	AS RQRD
		Terrain on ND	OFF
		Brake Fans	AS RQRD

♦ AFTER LANDING CHECKLIST	
RADAR & PRED WINDSHEAR	OFF

Parking Flow

PF		PM	
Parking Items			
Exterior Lights Runway Turnoff Lights, Nose Gear Lights	OFF		
Parking Brake ACCU Pressure (green)	**CHECKED**		
Parking Brake	**SET ON**		
Brake Pressure Indicators	**VERIFY SET**		
Anti-Ice System	CHECK **OFF**		
APU Bleed	**ON**		
Engine Master 1 & 2 Switches	**OFF**		
Yellow Electric Hydraulic Pump	**OFF**		
		PA: „Cabin Crew, all doos in Park."	PA ANNOUNCED
ECAM: Doors and Slides disarmed	CHECKED	ECAM: Doors and Slides disarmed	CROSSCHECKED
Beacon	**OFF**	**Temperature Selectors**	**OFF**
Seat Belt Signs	OFF	**Oxygen Crew Supply**	**OFF**
ACARS: Arrival Message on AOC	RECEIVED	**Fuel Pumps**	**OFF**
MCDU Data: Position Monitor	CHECK IRS	**Wing Lights**	CHECK **OFF**
		Transponder	**STBY**
„Parking Checklist"	ANNOUNCE		
Parking Checklist	READ	Parking Checklist	READ
Ground Contact	ESTABLISHED	**External Power**	**ON**
Cockpit Door	UNLOCKED	**APU Bleed**	**OFF**
		APU Master Switch	**OFF**
		Elapsed Time (Block Time)	NOTICED
		Fuel Quantity	CHECKED

♦ PARKING CHECKLIST	
PARKING BRAKE or CHOCKS	SET
ENGINES	OFF
FUEL PUMPS	OFF
YELLOW ELEC PUMP	OFF

Securing the Aircraft Procedure

Secure Flow is only started after the last passenger has left the cabin if no turnaround is planned but the aircraft is to be parked:

PF		PM	
Securing the Aircraft			
Parking Brake	CHECK ON		
ADIRS 1, 2 and 3 Mode Selectors	OFF		
ECAM: Maintenance Status	CHECKED		
ATC	STANDBY		
Brake Fan	OFF		
		Oxygen Crew Supply	OFF
		Exterior Lights (NAV Lights)	OFF
		No Smoking Signs	OFF
		Emergency Exit Lights	OFF
		APU Bleed	OFF
		APU Master Switch	OFF
		External Power	AS RQRD
		Battery 1 & 2	OFF
		Windows	CLOSED & LOCKED

♦ SECURING THE AIRCRAFT	
OXYGEN	OFF
EMERGENCY EXIT LTS	OFF
EFBs	OFF
BATTERIES	OFF

Rejected Takeoff Procedure

CM1		CM2	
Rejected Takeoff Procedure			
Decision before V_1			
RTO	ANNOUCE „STOP“		
Thrust Levers	**IDLE**		
MAX available Reverse Thrust	**SET**	Reversers: „Reverse Green”	CHECK, ANNOUNCE
		Deceleration	CHECK, ANNOUNCE
		Any Audio Warning	CANCEL
when Aircraft stopped			
Reverse Thrust	STOWED	Advice ATC	DONE
Parking Brake	APPLY	EMER EVAC Procedure	PREPARE
Cabin Crew (PA)	ALERT		
Oder: “**ECAM Actions**”	ANNOUNCED	Achieve **ECAM Actions**	PERFORM

CM1		CM2
EMER EVAC Procedure		
Aircraft / Parking Brake	STOP / ON	
ATC (on VHF1)	NOTIFY	
Cabin Crew (PA)	ALERT	
ΔP	CHECK ZERO	
ALL Engine Master Switches	OFF	
ALL Fire Pushbuttons (ENG & APU)	PUSH	
ALL Agents (ENG and APU)	AS RQRD	
if Evacuation required		
EVACUATION	INITIATE	
if Evacuation not required		
Cabin Crew and Passengers (PA)	NOTIFY	

Indications:

- **<u>Low Speed Regime (< 100kt):</u>**

  Here, pilots should tend to be “**stop-minded**”, which means that an RTO should be considered **for every ECAM warning or caution**.

- **<u>High Speed Regime (> 100kt):</u>**

  The closer the aircraft approaches the calculated V_1, the more “**go-minded**” one should be and an RTO should only be considered in the event of major failure, sudden loss of thrust or for indications that a flight cannot be conducted safely. This means the following **indicators for a Rejected Takeoff** (**RTO**) **in the high-speed range above 100kt before V_1**:

  1. **Fire Warning or Severe Engine Damage**
  2. **Sudden Loss of Engine Thrust** (as a rule Engine Flameout)

  } **= Engine Fail**

  3. **Aircraft unable or unsafe to fly** (Malfunctions or conditions that are unquestionable indications that the aircraft will not be able to fly safely.)
  4. **If any ECAM-Alert is triggered**. Up to 1,500ft AGL all ECAM cautions are suppressed, unless they are errors that are critical in take-off (e.g. ENG REVERSER FAULT, ENG REVERSE UNLOCKED, ENG 1(2) THR LEVER FAULT, ENG FAIL, F/CTL SIDESTICK FAULT). So if an error gets through, this immediately and automatically means a rejected take-off.

  In the event of a ruptured tire within a range of 20kt to V_1, it is recommended to still perform the takeoff, reduce the aircraft's fuel weight and then land safely using the full available runway length.

In addition to speed as a decision criterion, the **runway condition** also plays a not insignificant role in the decision-making process. A mindset is recommended, whereby you should be more **stop-minded** on a dry runway and more go-minded on a wet runway.

Cold Weather Operations

Icing Conditions:

- If the **OAT** (outside air temperature) on the ground or after takeoff or the **TAT** (true air temperature) during the flight is **+10°C** to a **static air temperature** (SAT) of **-40°C**.

and additionally

- **visible moisture** is present in any form. These include:
  - clouds,
  - fog with a visibility of 1.800m or less,
  - rain,
  - snow,
  - sleet, freezing rain and
  - visible ice crystals.

This also applies **on the ground** if:

- there is **snow**, **slush** or **standing water** on the ground surface on the apron, taxiways and/or runway.

Engine-Anti-Ice:

The Engine Anti Ice is always switched on during the flight if the **TAT or OAT is below +10°C and** at the same time the **SAT (static air temperature) indicates a warmer temperature than -40°C when flying through heavy haze** (visible moisture) or clouds, as well as through all types of precipitation.

Wing-Anti-Ice:

The **Wing Anti Ice** in the Airbus is **only used as required when ice forms**, as it draws massive amounts of performance. There is an ice detector (**Ice Indicator**) outside in the middle between the cockpit windows. If ice builds up there, the Wing Anti Ice is switched on, as it can then be assumed that the leading edge devices on the wings are also iced up. As soon as the Ice Indicator is free of ice deposits, the Wing Anti Ice can be **switched off again**. The Ice Indicator can also be **illuminated** to make it easier to see.

Probe/Window Heat:

Airbus recommends switching on the **PROBE/WINDOW HEAT switch** on the overhead panel at an **OAT** (**TAT**) **of ≤ 0°C during preliminary cockpit preparation**. **As soon as the two engines are started**, the PROBE/**WINDOW HEAT switch** is **switched off again**, as this system is then automatically adopted and operated.

After Landing:

Under **icing conditions** or if there is **contamination** such as snow or slush on the edges of the runway and/or taxiways, the **flaps are not retracted** after landing and therefore **remain in Config FULL or 3**, depending on the landing, until the **ground crew** can **confirm** at the gate that **no icing** has occurred in the area of the flaps. If the flaps are clear, they can be retracted, **otherwise** they must first be **de-iced**.

De-/Anti-Icing:

Ground Communication	
Cockpit	**Deicing Coordinator („Iceman“)**
	„AUAxxxx from Iceman 4“
„Iceman 4 from AUAxxxx“	
	„AUAxxxx my recommendation is: OAT xx°C 2-step procedure with Type 1 and Type 4 100% For wings and stabilizers, do you agree? And please state your wing fuel temperature“
„AUAxxxx Fuel temperature equals OAT and we accept your recommendation“	
	„AUAxxxx is the aircraft ready for deicing?“

Pilot Flying (PF)		Pilot Monitoring (PM)	
Airframe Deicing / Antiicing Procedure on Ground			
Parking Brake	ON		
ECAM Fuel Page: Lowest Fuel Temperature	REPORTED		
Cabin Pressure Mode Selector	**AUTO**		
Engine 1 Bleed	**OFF**		
Engine 2 Bleed	**OFF**		
APU Bleed	**OFF**		
Ditching Pushbutton	**ON**		
Thrust Levers	**CHECK IDLE**		
ECAM Pressure Page: Check Outflow Valve closed	CHECKED	Inform Ground Crew: „Aircraft prepared for spraying“	ANNOUNCED

Ground Communication	
Cockpit	**Deicing Coordinator („Iceman“)**
	„AUAxxxx we start with the de-icing now and call you when we’re finished.“
When De-Icing is finished	
	„AUAxxxx we have completed the de-/anti-icing with Type 1 followed by Type 4 for wings and stabilizers. Your Anti-icing code is: Type 4 100% at 1300 local time. Post de-/anti-icing check is performed.“
„AUAxxxx confirm, wings and stabilizers de-iced. Anti-Icing Code: Type 4 100% 1300 local time.“	
	„AUAxxxx from Iceman 4, I’ll give you the “All Clear Sign” from the right side. Good Bye.“
„AUAxxxx, All Clear Sign from the right. Thank you. Good bye.“	

Pilot Flying (PF)		Pilot Monitoring (PM)	
Airframe Deicing / Antiicing Completed Procedure on Ground			
De-Icing / Anti-Icing Report	RECEIVED		
Ditching Pushbutton	**OFF**		
ECAM Pressure Page: Check Outflow Valve open	CHECKED		
Engine 1 Bleed	**ON**		
Engine 2 Bleed	**ON**		
Ground Equipment	REMOVED		
Flight Controls[A]	**CHECKED**	**Flight Controls**[A]	**CHECKED**
		Flaps[A]	**SET FOR T/O**

[A]) In the event of active precipitation, the flight controls check and the setting of the flaps for take-off may only take place at the runway holding point!

Fluid-Types:

Fluid Type	Farbe	Zusammensetzung	Holdover Time	min. Rotation Speed
Type I	rot-orange	heißes Wasser etwas Alkohol	6 – 11min.	kein Minimum
Type II	durchsichtig bis strohgelb	50% heißes Wasser 50% Alkohol	20 – 45min.	100kt
Type III	gelb-grün	75% Alkohol 25% heißes Wasser Verdickungsmittel	10 – 20min.	60kt
Type IV	smaragdgrün	Alkohol etwas heißes Wasser Verdickungsmittel	35 – 75min.	100kt

Clean Aircraft Concept!

- **Single Step Procedure**:
  - De-Icing with Fluid Typ I; Holdover-Time starts at the begin of spraying of Type I (set Elapsed Timer!)
- **Two-Step-Procedure**:
  - De-Icing with Fluid Typ I
  - Thereafter Anti-Icing with Fluid Type III or IV, Holdover-Time starts at the begin of spraying of Type III or IV (set Elapsed Timer!)

Holdover-Time (**HOT**):

Fluid Type I

OAT		Approximate holdover times under various weather conditions (hours: minutes)						
°C	°F	Frost (*)	Freezing Fog	Snow	Freezing Drizzle (**)	Light Freezing Rain	Rain on Cold Soaked Wing	Others (1)
above 0	above 32	0:45	0:12-0:30	0:06-0:15	0:05-0:08	0:02-0:05	0:02-0:05	CAUTION: No holdover time guidelines exist
0 to -10	32 to 14	0:45	0:06-0:15	0:06-0:15	0:05-0:08	0:02-0:05	CAUTION: Clear ice may require touch for confirmation	
below -10	below 14	0:45	0:06-0:15	0:06-0:15				

Fluid Tyie III

OAT		SAE Type IV fluid concentration Neat-fluid/water (Vol. %/Vol. %)	Approximate holdover times under various weather conditions (hours: minutes)						
°C	°F		Frost (*)	Freezing fog	Snow	Freezing drizzle (***)	Light freezing rain	Rain on cold soaked wing	Others (1)
above 0	above 32	100/0	18:00	1:05-2:15	0:35-1:05	0:40-1:00	0:25-0:40	0:10-0:50	CAUTION: No holdover time guidelines exist
		75/25	6:00	1:05-1:45	0:20-0:40	0:30-1:00	0:15-0:30	0:05-0:35	
		50/50	4:00	0:20-0:35	0:05-0:20	0:10-0:20	0:05-0:10	CAUTION: Clear ice may require touch for confirmation	
0 to -3	32 to 27	100/0	12:00	1:05-2:15	0:30-0:55	0:40-1:00	0:25-0:40		
		75/25	5:00	1:05-1:45	0:20-0:35	0:30-1:00	0:15-0:30		
		50/50	3:00	0:20-0:35	0:05-0:15	0:10-0:20	0:05-0:10		
below -3 to -14	below 27 to 7	100/0	12:00	0:40-1:30	0:20-0:40	0:20-0:55 (**)	0:10-0:30 (**)		
		75/25	5:00	0:25-1:00	0:15-0:25	0:20-0:55 (**)	0:10-0:30 (**)		
below -14 to -25	below 7 to -13	100/0	12:00	0:20-0:40	0:15-0:30				
below -25	below -13	100/0	SAE Type IV fluid may be used below -25°C (-13°F) provided the freezing point of the fluid is at least 7°C (13°F) below the OAT and the aerodynamic acceptance criteria are met. Consider use of SAE Type I when SAE Type IV fluid cannot be used.						

Fluid Type IV

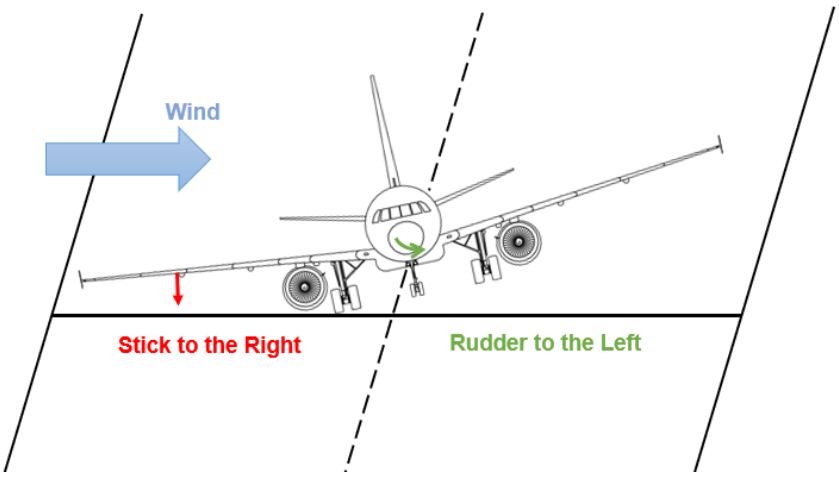
OAT		SAE Type IV fluid concentration Neat-fluid/water (Vol. %/Vol. %)	Approximate holdover times under various weather conditions (hours: minutes)						
°C	°F		Frost (*)	Freezing fog	Snow	Freezing drizzle (***)	Light freezing rain	Rain on cold soaked wing	Others (1)
above 0	above 32	100/0	18:00	1:05-2:15	0:35-1:05	0:40-1:00	0:25-0:40	0:10-0:50	CAUTION: No holdover time guidelines exist
		75/25	6:00	1:05-1:45	0:20-0:40	0:30-1:00	0:15-0:30	0:05-0:35	
		50/50	4:00	0:20-0:35	0:05-0:20	0:10-0:20	0:05-0:10	CAUTION: Clear ice may require touch for confirmation	
0 to -3	32 to 27	100/0	12:00	1:05-2:15	0:30-0:55	0:40-1:00	0:25-0:40		
		75/25	5:00	1:05-1:45	0:20-0:35	0:30-1:00	0:15-0:30		
		50/50	3:00	0:20-0:35	0:05-0:15	0:10-0:20	0:05-0:10		
below -3 to -14	below 27 to 7	100/0	12:00	0:40-1:30	0:20-0:40	0:20-0:55 (**)	0:10-0:30 (**)		
		75/25	5:00	0:25-1:00	0:15-0:25	0:20-0:55 (**)	0:10-0:30 (**)		
below -14 to -25	below 7 to -13	100/0	12:00	0:20-0:40	0:15-0:30				
below -25	below -13	100/0	SAE Type IV fluid may be used below -25°C (-13°F) provided the freezing point of the fluid is at least 7°C (13°F) below the OAT and the aerodynamic acceptance criteria are met. Consider use of SAE Type I when SAE Type IV fluid cannot be used.						

Runway Condition Assessment Matrix (**RCAM**) for Contaminated Runway:

Runway Surface Conditions		Observations on Deceleration and Directional Control	Related Landing Performance		Maximum Crosswind for Landing (Gust included)
Runway State or / and Runway Contaminant	AIREP[1]		RWYCC[2]	Level	
Dry	-	-	6	DRY	38 kt
Damp **Wet** Up to 3 mm (1/8") of water **Slush** Up to 3 mm (1/8") **Dry snow** Up to 3 mm (1/8") **Wet snow** Up to 3 mm (1/8") **Frost**	Good	Braking deceleration is normal for the wheel braking effort applied. Directional control is normal.	5	GOOD	38 kt
Compacted snow OAT at or below -15 °C	Good to Medium	Braking deceleration and controllability is between Good and Medium.	4	GOOD TO MEDIUM	29 kt
Dry snow More than 3 mm (1/8"), up to 100 mm (4") **Wet snow** More than 3 mm (1/8"), up to 30 mm (6/5") **Compacted snow** OAT above -15 °C **Dry snow over compacted snow** **Wet snow over compacted snow** **Slippery wet**	Medium	Braking deceleration is noticeably reduced for the wheel braking effort applied. Directional control may be reduced.	3	MEDIUM	25 kt
Standing Water More than 3 mm (1/8"), up to 13 mm (1/2") **Slush** More than 3 mm (1/8"), up to 13 mm (1/2")	Medium to Poor	Braking deceleration and controllability is between Medium and Poor. Potential for hydroplaning exists.	2	MEDIUM TO POOR	20 kt
Ice (cold & dry)	Poor	Braking deceleration is significantly reduced for the wheel braking effort applied. Directional control may be significantly reduced.	1	POOR	15 kt
Wet ice **Water on top of Compacted Snow** **Dry Snow or Wet Snow over ice**	Less than Poor	Braking deceleration is minimal to non-existant for the wheel braking effort applied. Directional control may be uncertain.	-	-	-

Crosswind Landing Technique

- In general, a crosswind landing in an Airbus is permitted up to a maximum crosswind of 35kts including gusts. The worse the runway conditions are, the less the maximum crosswind is allowed. For slush and frost, a maximum crosswind of 25kts is permitted and if the runway is icy, a landing is only permitted with a crosswind of 5kts.
- The **outer wing**, i.e. the one **in the direction of the wind**, is **pushed down** with the **sidestick** (**aileron**). At the **same time**, the **heading** is **aligned with the runway track** using the **rudder** (**pedals**), i.e. stepped **against the wind direction and thus deflected**.
  The aim is to touch down on the runway first with the main gear facing the wind so that the wind cannot reach under the wing. The other main gear would then follow and finally the nose gear would touch down. The **principles** are as follows:
  - **Aileron into the Wind!**
  - **Rudder to avoid drift!**
  - **Little but not too much flare** → The goal is a good, solid touchdown!
  - Then at the same time **Thrust Idle**.



This technique is therefore also known as “slipping” or **cross-control**, as the rudder is stepped on the other side than the aileron (right stick - left rudder and vice versa).

Aviate	Navigate	Communicate
• **Autopilot ON**	• **at least MSA**	**Sequence**:
• **Fly the Aircraft**	• **Heading „clear“ of Obstacles** like mountains	**ATC** ↓
• **Normal Procedures** work off	• **„Where am I?“**	**Cabin Crew** ↓
	• **„Where do I want to go?“**	**Passagiere**
		→ **„Who needs to know what?“**

Decision making model FORDEC:

Facts
Options
Risks and Benefits
Decision
Execution
Check

NITS-Briefing (Cabin, ATC):

Nature
Intention
Time
Specials

Memory Items [MEM]

1. **Callout: „LOSS OF BREAKING“ [PRO-ABN-BRAKES]**

PF		PM	
[MEM] LOSS OF BREAKING			
If no Braking:			
Thrust Reverse	MAX		
Brake Pedals	RELEASE		
„Anti-Skid“ (Order)	OFF	Anti-Skid and Nose Wheel Steering	OFF
Brake Pedals	PRESS		
Maximum Brake Pressure	1.000psi		
If still no Braking:			
Parking Brake	USE[1]		

[1]) Use short and successive Applications of the Parking Brake to stop the Aircraft!

If necessary, Maximum Reverse Thrust can be applied until the aircraft comes to a complete stop.

2. **Callout: „UNRELIABLE AIRSPEED“ [PRO-ABN-NAV]**

PF		PM
[MEM] UNRELIABLE AIRSPEED INDICATION		
If the safe conduct of the Flight is impacted:[1]		
1. Knock Off the Automation		
Autopilot (AP)	OFF	
Autothrust (A/THR)	OFF	
Flight Directors	OFF	
2. Fly a Logical Attitude (Pitch / Thrust)		
• below THRUST RED ALT	15° / TOGA	
• below FL 100	10° / CLB	
• above FL 100	5° / CLB	
3. Get rid of the Drag		
FLAPS • if Config 0, 1, 2 or 3 • in Config FULL	 MAINTAIN FLAPS 3	
Speedbrakes	CHECK RETRACTED	
Landing Gear	UP	
4. Troubleshoot[2]		
LEVEL OFF	PERFORM	

[1]) If a safe execution of the flight is no longer guaranteed, which is the case if there is a loss of **situational awareness** or an inappropriate pitch and thrust situation in relation to the current flight phase and/or an unexpected flight path is flown under the current flight conditions, the memory items must be processed accordingly.

[2]) When at or above MSA or Circuit Altitude: Level-Off for Troubleshooting.

3. Callout: „WINDSHEAR, TOGA“ – Reactive Windshear Warning [PRO-ABN-SURV]

PF		PM
[MEM] WINDSHEAR[1]		
AT TAKEOFF before V1		
If there are significant variations in Airspeed and an Airspeed Trend Vector below the indicated V1[2]: Reject the Take-Off!		
AT TAKEOFF after V1		
Thrust Levers	**TOGA**	
Reaching VR	**ROTATE[3]**	
Pitch: **SRS Orders[4]**	**FOLLOW**	
WHEN AIRBORNE, INITIAL CLIMB OR LANDING		
Thrust Levers	**TOGA**	
Autopilot (AP)[5]	**KEEP ON**	
Pitch: **SRS Orders[4]**	**FOLLOW**	
Do NOT change Configuration (Slats, Flaps, Landing Gear) until Out of Windshear![6]		
Closely monitor Flight Path and Speed!		
Recover smoothly to normal Flight Path (Climb-Out) out of Windshear.		

[1]) Basically, you just have to remember the 5 steps here: **TOGA → SRS → CONFIG → FLIGHT PATH → RECOVERY**.

[2]) If the airspeed indications change significantly, the take-off is aborted before reaching V1 if the stop distance is still sufficient.

[3]) If necessary, the side stick can be fully retracted.

[4]) The SRS pitch specifications should be flown quickly and smoothly, but not aggressively. It may be necessary to pull the sidestick “full backward”. If the flight director bars are not displayed, an initial pitch attitude of 17.5° is recommended. The pitch can then be increased further if necessary to prevent a loss of altitude. In order to maximize the climb gradient, wings level should be flown unless a turn is necessary to avoid obstacles.

[5]) As part of alpha floor protection, an automatic disconnect of the autopilot system can occur if the angle of attack exceeds the α-Prot. value. If the autopilot is not (yet) engaged at the time the windshear warning occurs, it will not be activated now either.

[6]) The aircraft configuration should not be changed under any circumstances until you are out of the windshear conditions, as retracting or extending landing gear, flaps or slats, as well as opening the landing gear doors, leads to a further increase in drag.

4. Callout: „WINDSHEAR AHEAD“ – Predictive Windshear Warning [PRO-ABN-SURV]

PF		PM
[MEM] WINDSHEAR AHEAD		
AT TAKEOFF during the Take-Off Run		
Reject the Take-Off!		
WHEN AIRBORNE		
Thrust Levers	**TOGA**	
Autopilot (AP; if engaged)	**KEEP ON**	
Pitch: **SRS Orders[4]**	**FOLLOW**	
AT LANDING		
Go-Around	**PERFORM**	
Autopilot (AP; if engaged)	**KEEP ON**	

There is a possibility of false positive predictive windshear warnings. A windshear warning can be ignored if:

- other signs of possible windshear conditions are missing and
- the reactive windshear system is switched on and active.

However, you must always rely on reactive windshear alerts and react accordingly! It is generally advisable to delay the take-off for the departure or to select the most favorable runway for the take-off in terms of wind conditions. The sltas/flaps configuration can also be changed as long as you have not yet flown into the windshear.

5. **Callout: „PULL UP, TOGA“ – EGPWS Warnings [PRO-ABN-SURV]**

PF		PM
[MEM] PULL UP, TOGA (EGPWS WARNING – Terrain Avoidance Maneuver)		
„PULL UP” – „TERRAIN AHEAD, PULL UP” – „OBSTACLE AHEAD, PULL UP”		
Autopilot (AP)	**OFF**	
Pitch	**PULL UP**	
Thrust Levers	**TOGA**	
Speedbrakes Lever	**CHECK RETRACTED**	
Bank Angle	**WINGS LEVEL or ADJUST**	
Do not change Configuration (Slats, Flaps, Landing Gear) until clear of Obstacle.		

The five memory items of the terrain avoidance maneuver should be executed simultaneously, i.e. all at the same time if possible. The aim is to increase the pitch with “**full backstick**”, i.e. full pull backwards on the sidestick, while keeping the wings horizontal (“**keep wings level**”) and keeping the sidestick in this position.
A turning maneuver can be initiated if this is the safest action. In this case, the PULL-UP maneuver must be executed before the turn.

6. **Callout: VARIOUS – EGPWS Cautions [PRO-ABN-SURV]**

PF	PM
[MEM] EGPWS CAUTIONS	
„TERRAIN TERRAIN” – „TOO LOW, TERRAIN”	
During night or in IMC: **Terrain Avoidance Maneuver**. During Daylight and in VMC, with Terrain and Obstacles clearly in sight: **Adjust the Flight Path** or initiate a Go-Around.	
„TERRAIN AHEAD” – „OBSTACLE AHEAD”	
Adjust the Flight Path. **Stop Descent**. Climb and/or turn, as necessary, based on analysis of all available instruments and information.	
„SINK RATE” – „DON’T SINK”	
Adjust Pitch Attitude and Thrust to correct and **adjust the Flight-Path** and therefore to silence the alert. When below 1.000ft AAL (above Aerodrome Level) in IMC or below 500ft in VMC, **consider** conducting a **Go-Around**.	
„TOO LOW, GEAR” – „TOO LOW, FLAPS”	
Perform a **Go-Around**.	
„GLIDESLOPE”	
Adjust the Flight-Path to establish the Aircraft on the Glideslope or set the G/S MODE Pushbutton to „OFF”, if flight below the glideslope is intentional (non precision approach). When below 1.000ft AAL (above Aerodrome Level) in IMC or below 500ft in VMC, **consider** conducting a **Go-Around**.	

7. Callout: „TCAS, I HAVE CONTROL“ – TCAS Warnings [PRO-ABN-SURV]

PF		PM	
[MEM] TCAS Alert – I HAVE CONTROL			
Traffic Alert (TA): „TRAFFIC, TRAFFIC”			
„TCAS – I have control“	ANNOUNCE		
Awareness, but don’t perform any evasive maneuver (Sidestick in Hand)			
FMA: „TCAS blue“	ANNOUNCE		
Resolution Advisory (RA): „CLIMB, CLIMB”, „DESCENT, DESCENT”[1]			
Autopilot (AP; if engaged)	**OFF**		
Flight Directors (FD; BOTH!)	**OFF**		
Respond promptly and smoothly.			
Pitch (Vertical Speed): adjust or maintain	**VSI GREEN**		
		ATC	NOTIFY
for any CLIMB RA during Approach in CONF 3 or FULL			
Go-Around	**PERFORM**		
SRS GA Mode	**FOLLOW**		
Vertical Speed	**MONITOR**		
		ATC	NOTIFY
when „CLEAR OF CONFLICT“			
Normal Navigation	**RESUME**	**ATC**	**NOTIFY**
Lateral and Vertical Guidance	**ADJUST**		
Flight Directors	**RE-ENGAGE**		
Autopilot	**RE-ENGAGE**		

[1]) This includes all messages of the type „CLIMB” and „DESCEND” or „MAINTAIN VERTICAL SPEED MAINTAIN” or „ADJUST VERTICAL SPEED ADJUST” or „MONITOR VERTICAL SPEED”.

8. Callout: „EMERGENCY DESCENT“ [PRO-ABN-MISC]

PF		PM
[MEM] EMERGENCY DESCENT		
CREW OXY MASKS[1]	**USE**	
Signs **(Seat Belt, No Smoking, EMER EXT Lights)**	**ON**	
EMERGENCY DESCENT	**INITIATE**	
• FCU: **ALT** Selector (select lower Altitude)	**TURN and PULL**	
• FCU: **HDG** Selector (~30° away from Airway)	**TURN and PULL**	
• FCU: **Speed** Selector (maintain present Speed)	**PULL**	
if Autothrust (A/THR) not on:		
• **Thrust-Levers**	**IDLE**	
Speedbrake	**FULL**	

[1]) To save oxygen, the Oxygen Diluter Selector should be set to the N-position.

9. Callout: „STALL, I HAVE CONTROL“ [PRO-ABN-MISC]

PF		PM
[MEM] STALL RECOVERY		
Nose Down Pitch Control[1] to reduce AOA	APPLY	
Bank Angle	WINGS LEVEL	
when out of Stall Conditions (no longer Stall Indications)		
Thrust:[2]	INCREASE SMOOTHLY	
Speedbrakes	CHECK RETRACTED	
Flight Path	RECOVER SMOOTHLY	
if in Clean Configuration and below 20.000ft		
FLAPS 1	SELECT	

[1]) The **nose pitch down moment** on the sidestick means a “**push to unload**”, i.e. to **immediately lower the nose**, which immediately **reduces the angle of attack** at the same moment. This is the direct and immediate key action to reduce the AOA and thus lead to a recovery of lift on the wings.
If the forward pressure on the sidestick alone does not cause the pitch to drop sufficiently or fast enough, it may be necessary to **additionally reduce the thrust**.

[2]) In the case of a **single-engine operation**, the thrust asymmetry is compensated or equalized with the **rudder** while the thrust is increased.

10. Callout: „STALL, TOGA 15°“ – STALL WARNING AT LIFT-OFF [PRO-ABN-MISC]

PF		PM
[MEM] STALL WARNING AT LIFT-OFF		
Thrust	TOGA	
At the same time:		
Pitch Attitude	15°	
Bank Angle	WINGS LEVEL	

A safe flight path and a safe speed should be achieved by using the stall warning procedure. If the stall warning still persists, it is to be regarded as a false positive. This can occur in normal law if the AOA probe (Angle of Attack Sensor) is damaged.